

Lecture 10

Representation Learning and Information Theory

Speaker: Kaiming He

Overview

- Basics of Information Theory
- Modeling Information with Neural Networks

What are “good” representations?

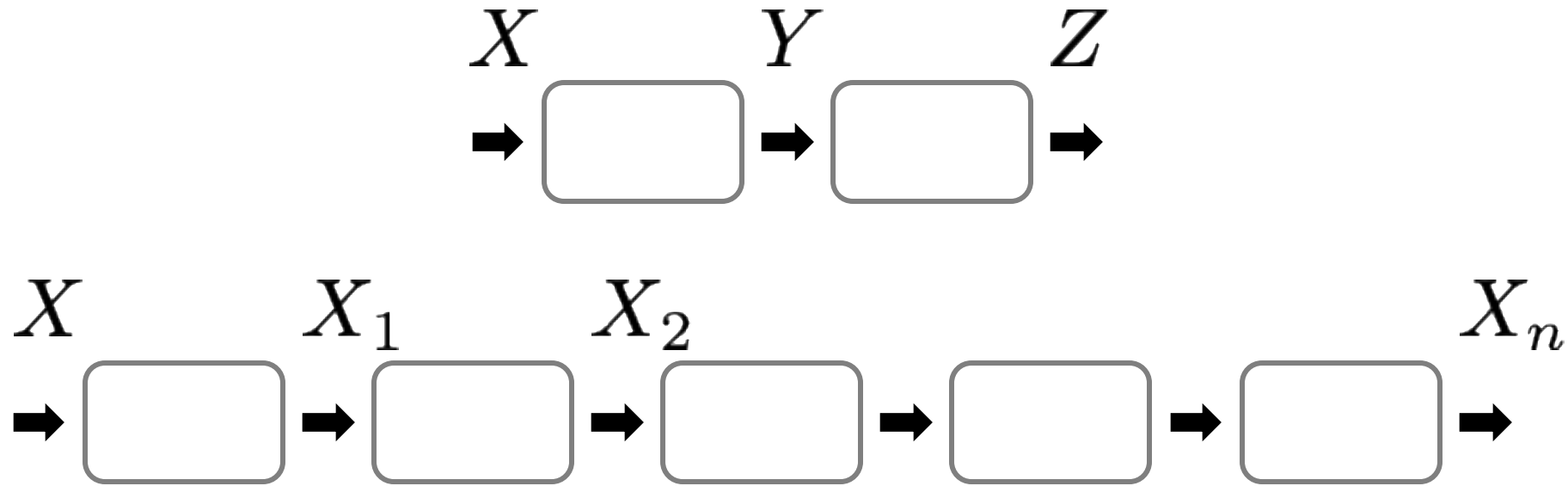
We may want representations that:

- retain information about the input
- discard information irrelevant to the tasks
- compress information effectively under limited capacity

But:

- What is “information”?
- How can we measure information?
- with neural networks?

In the context of neural nets:



We may ask questions such as:

- What is the mutual information between X and Z ?
- How much information is lost in X_n vs. X_2 ?
- Is X_n more informative than Z ?
- ...

Basics of Information Theory

See also:
6.7480 Information Theory

Information Measures

- **Entropy**: measure of uncertainty

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x) = -\mathbb{E}_{x \sim p} \log p(x)$$

X : a random discrete variable

$\mathcal{X}$: the support set e.g. $\mathcal{X} = \{0, 1, 2, 3\}$

x : a realization e.g. $x = 2$

Information Measures

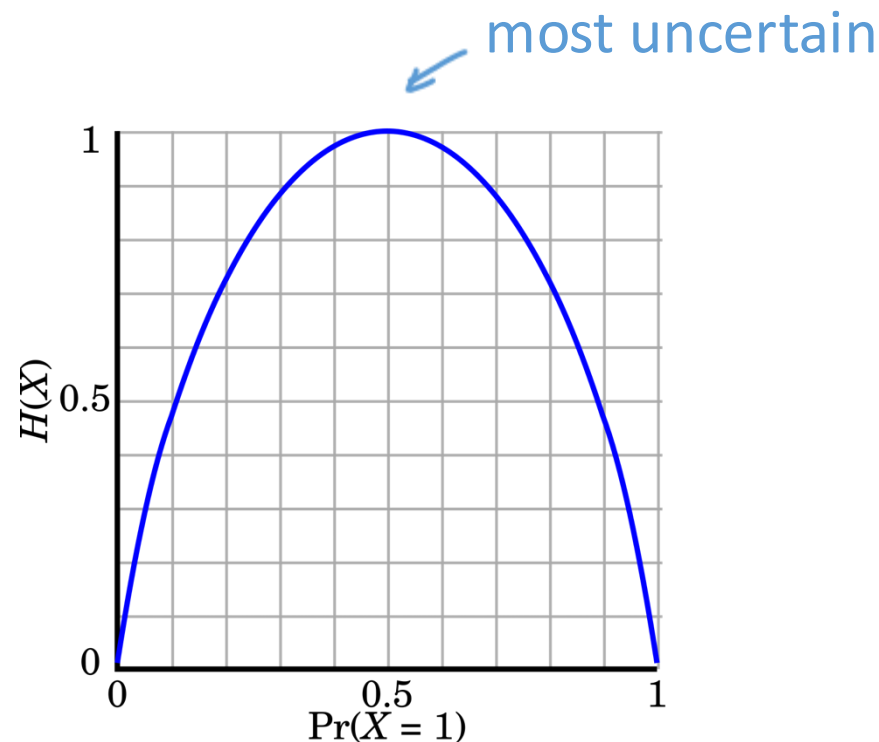
- **Entropy:** measure of uncertainty

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x) = -\mathbb{E}_{x \sim p} \log p(x)$$

Example: Binary entropy $\mathcal{X} = \{0, 1\}$

$$\text{Prob}(X=1) \triangleq p$$

$$H(X) = -p \log p - (1 - p) \log(1 - p)$$



Information Measures

- **Joint entropy:** uncertainty of joint distribution

$$H(X, Y) = -\mathbb{E}_{p(x,y)} \log p(x, y)$$

- **Conditional entropy:** uncertainty of X given Y

$$\begin{aligned} H(X \mid Y) &= \mathbb{E}_{p(y)} H(X \mid Y = y) \\ &= -\mathbb{E}_{p(x,y)} \log p(x \mid y) \end{aligned}$$

Information Measures

- **Mutual Information:** information provided by X about Y

$$I(X; Y) = \mathbb{E}_{p(x, y)} \log \frac{p(x, y)}{p(x)p(y)} = D_{\text{KL}}(p(x, y) \parallel p(x)p(y))$$

Properties:

$$I(X; Y) = I(Y; X)$$

symmetry

$$I(X; Y) = H(X) - H(X \mid Y)$$

reduction in uncertainty of X given Y

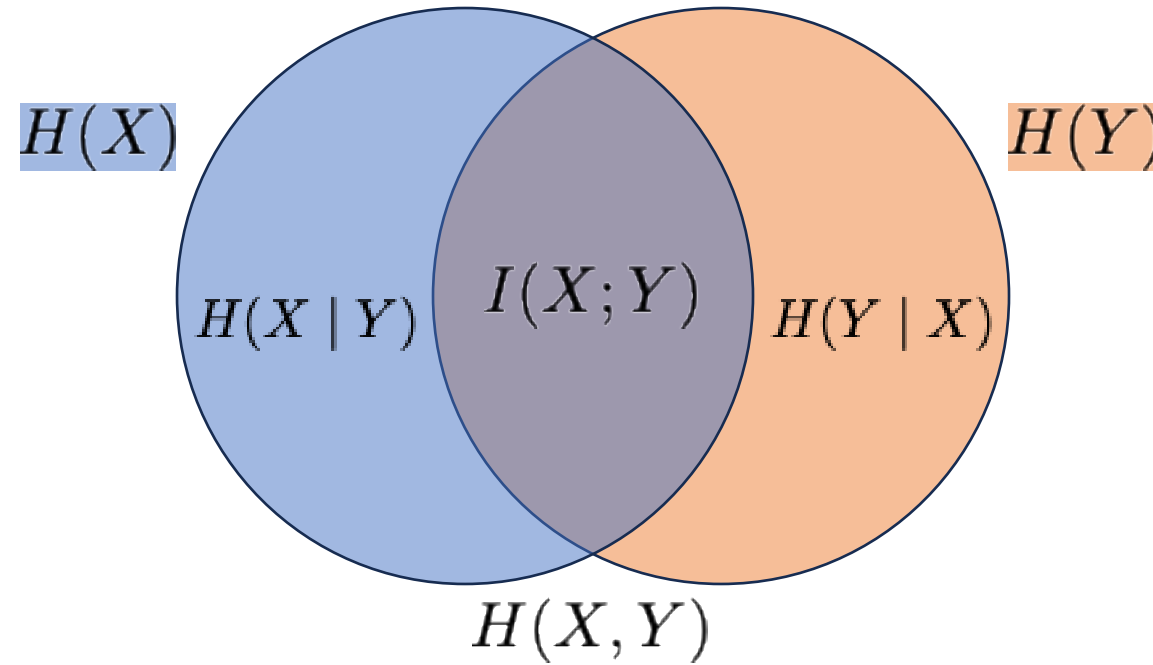
$$I(X; Y) = H(Y) - H(Y \mid X)$$

reduction in uncertainty of Y given X

Information Measures

$$I(X; Y) = H(X) - H(X | Y)$$

$$I(X; Y) = H(Y) - H(Y | X)$$

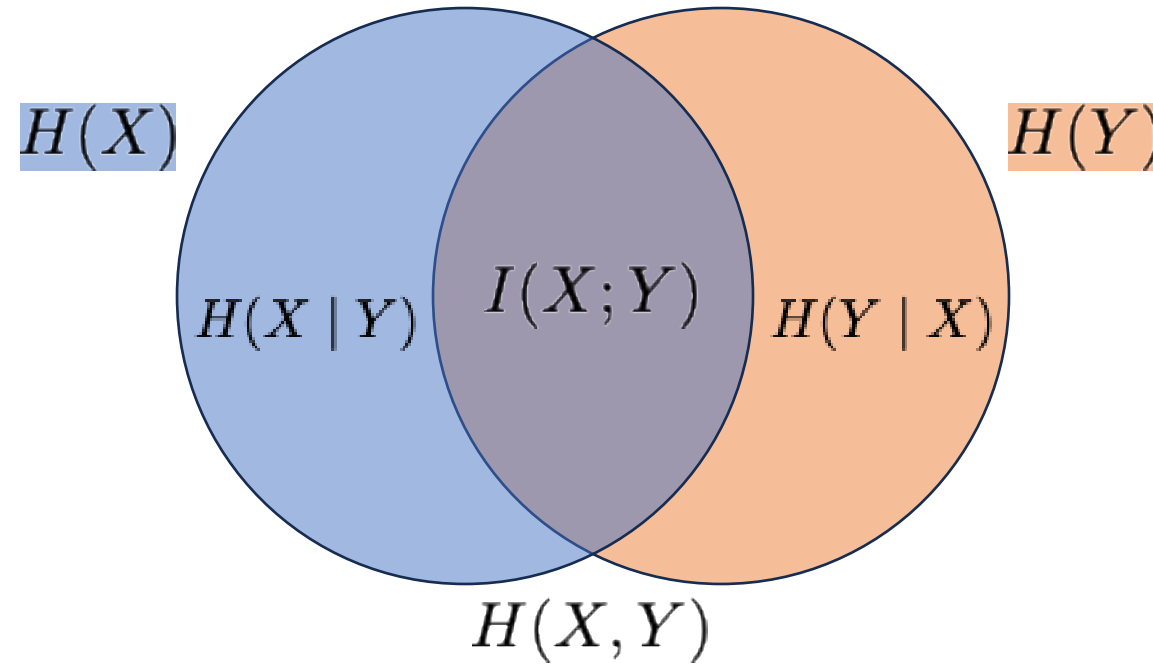


$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

Information Measures

$$I(X; Y) \leq H(X)$$

$$I(X; Y) \leq H(Y)$$

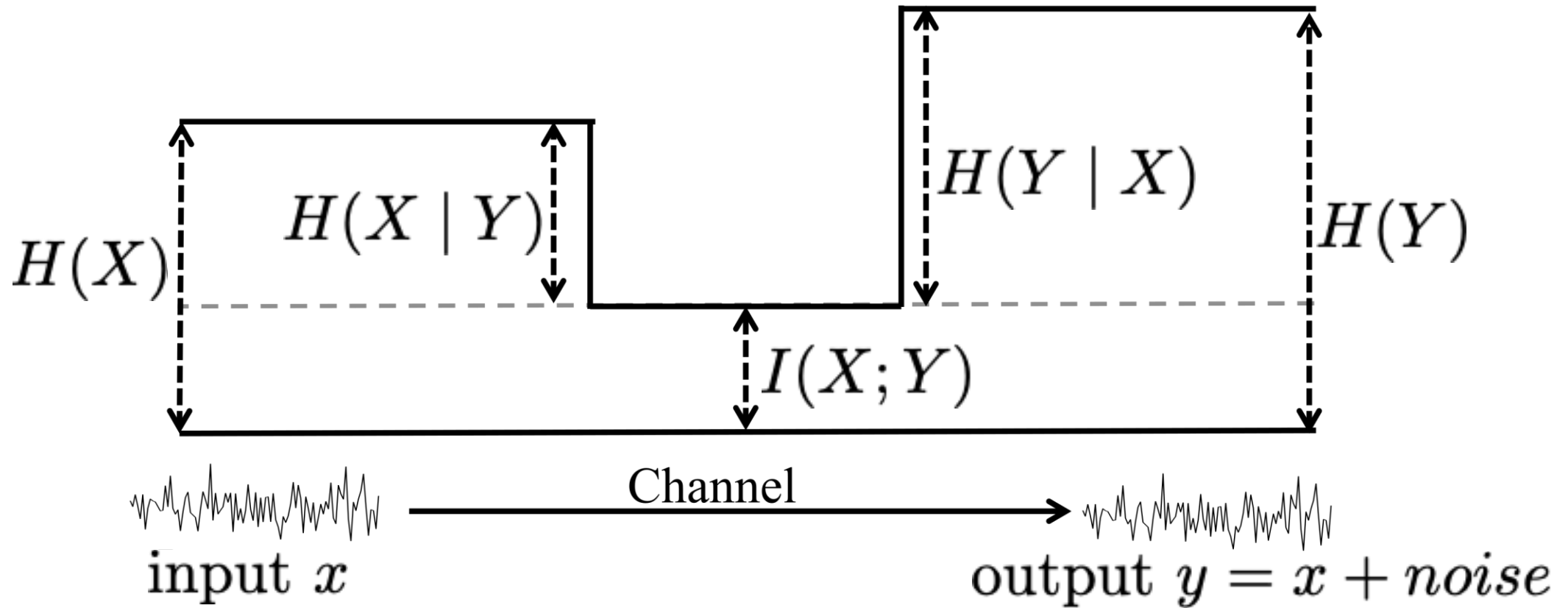


non-negative:

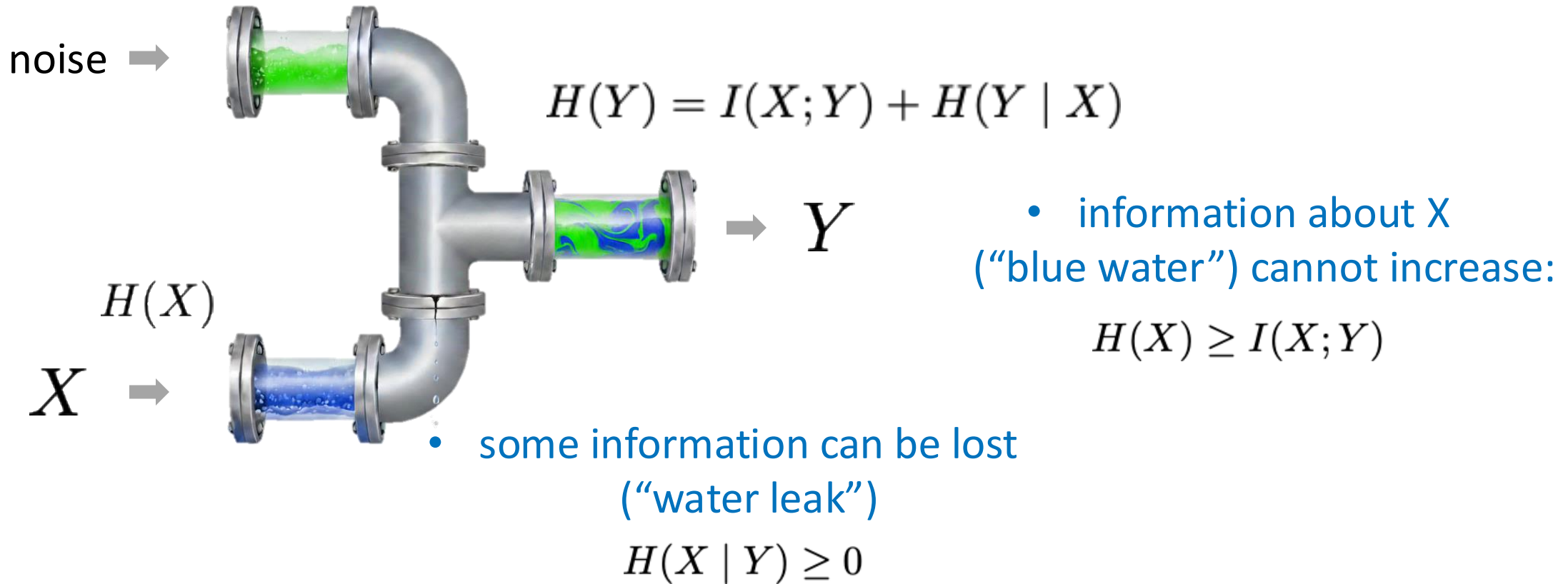
$$I(X; Y) \geq 0$$

$$I(X; Y) = 0 \iff X \perp Y$$

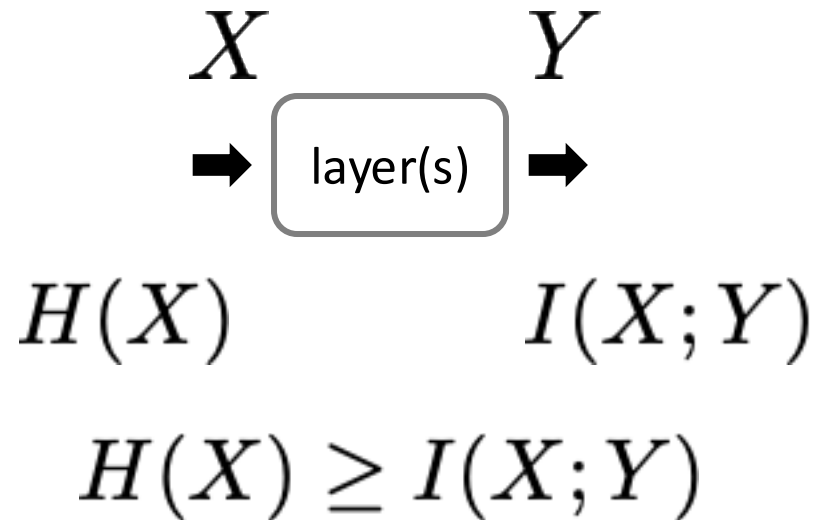
Information about X cannot increase through processing



Information about X cannot increase through processing



Information about X cannot increase through processing

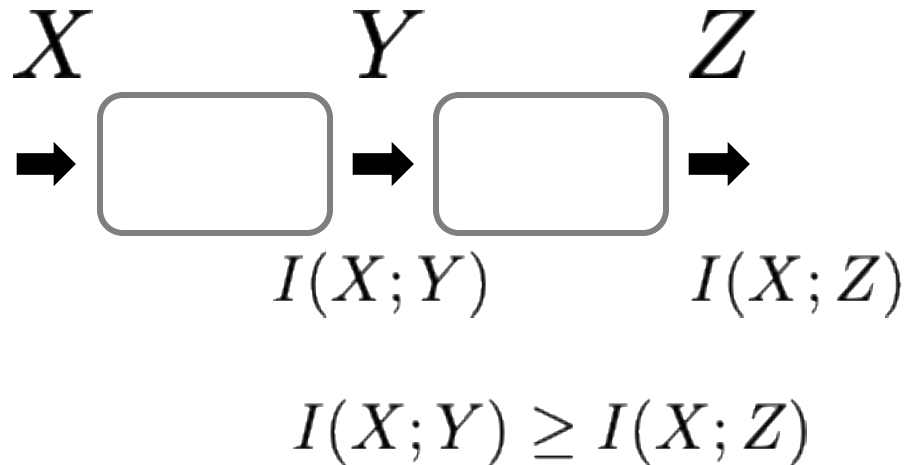


Any network layer(s) can only
reduce or keep information about its input

Data Processing Inequality

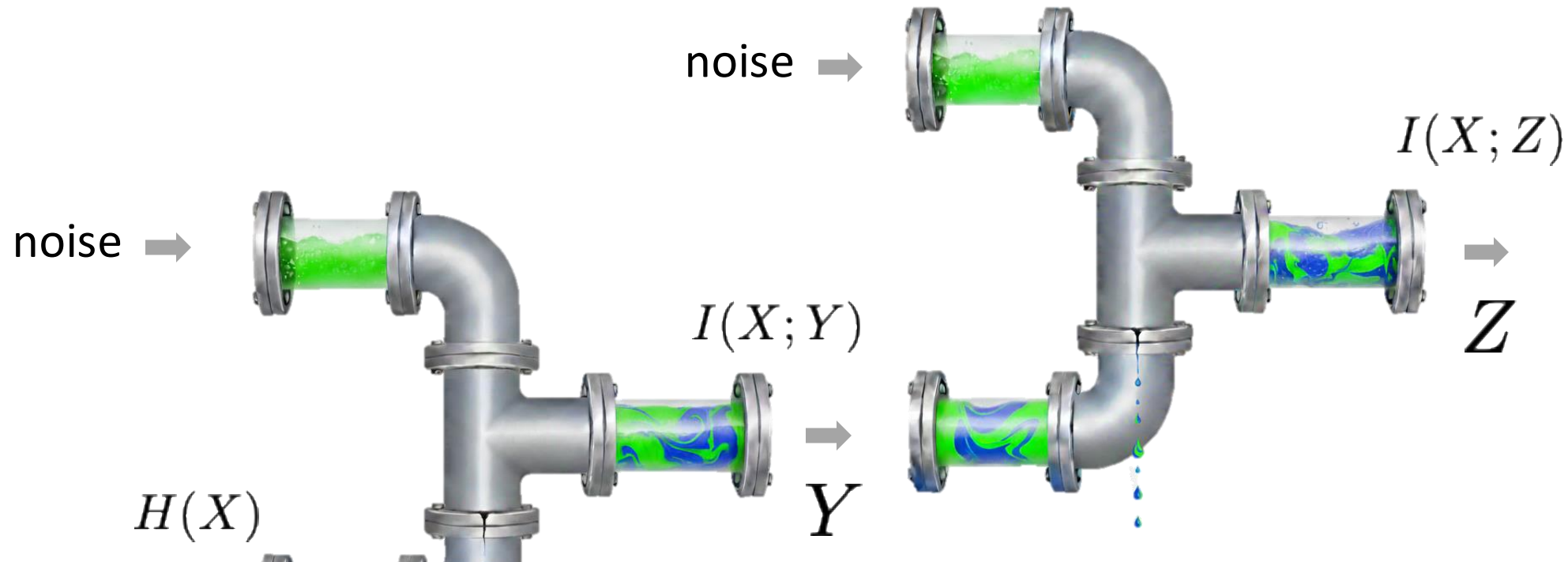
Consider a Markov Chain: $X \rightarrow Y \rightarrow Z$

we have: $I(X; Y) \geq I(X; Z)$



Information about the input degrades or stays the same with further processing.

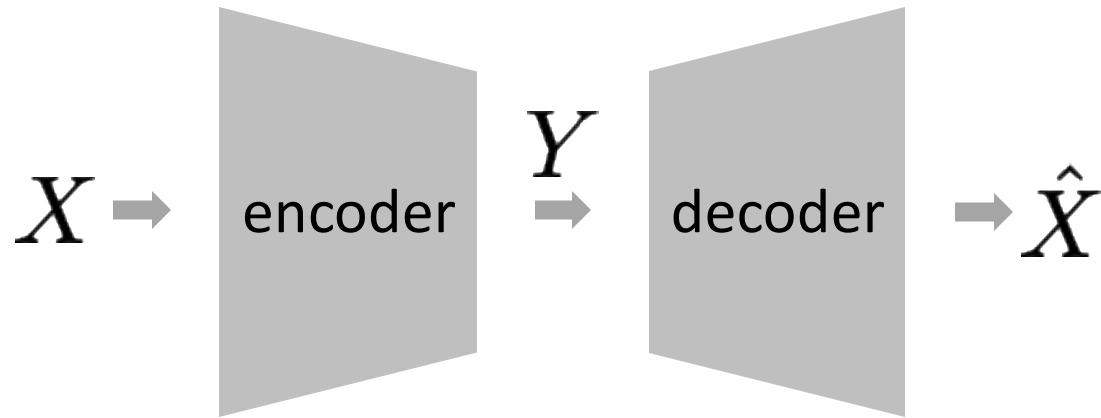
Data Processing Inequality



- information about X (“blue water”) cannot increase:

$$H(X) \geq I(X; Y) \geq I(X; Z)$$

Rate-Distortion Theory

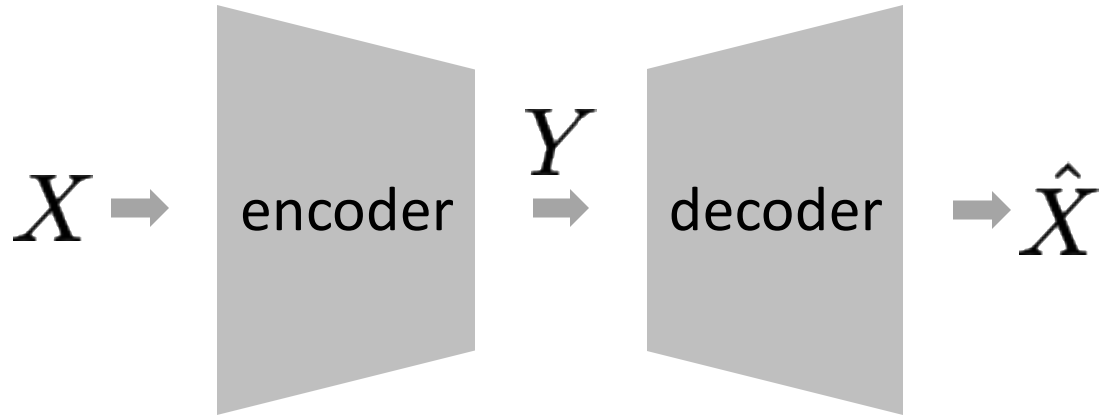


distortion measure: $d(x, \hat{x}) \geq 0$ e.g., Mean Squared Error (MSE)

rate: R average bits required to represent per realization

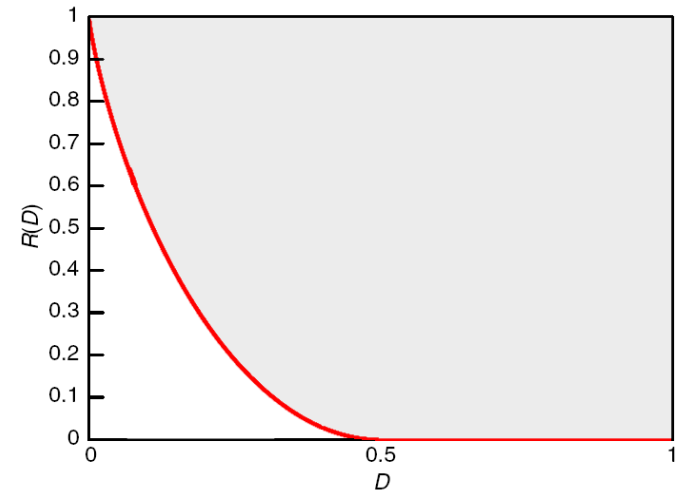
rate-distortion function: $R(D)$ minimum rate needed such that the expected distortion does not exceed D

Rate-Distortion Theory

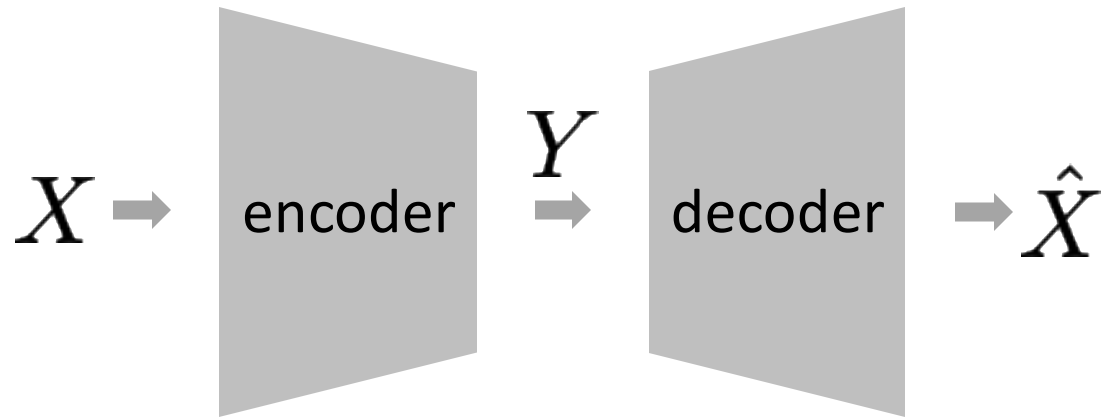


$$R(D) = \inf_{P(\hat{X}|X): \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X})$$

- lossless compression: $D = 0, R(D) = H(X)$
- As D increases, R decreases
- If D is infinite, R is 0



Rate-Distortion Theory

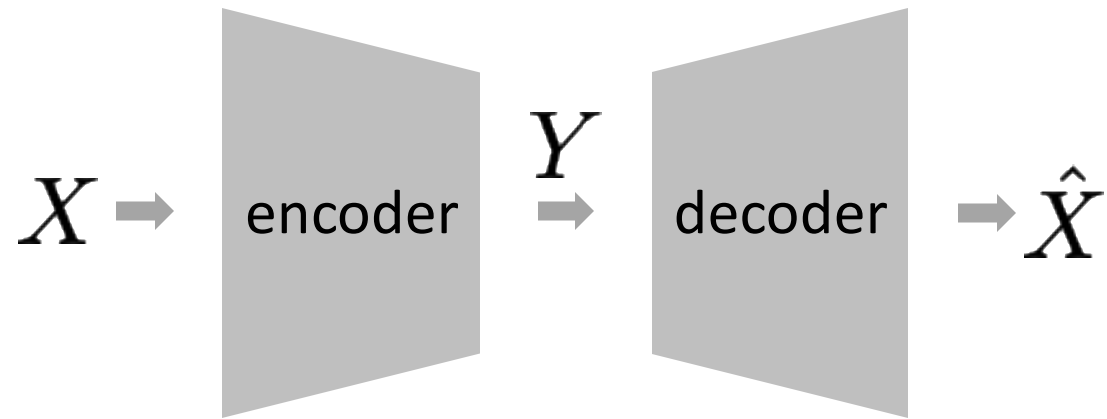


$$R(D) = \inf_{P(\hat{X}|X): \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X})$$

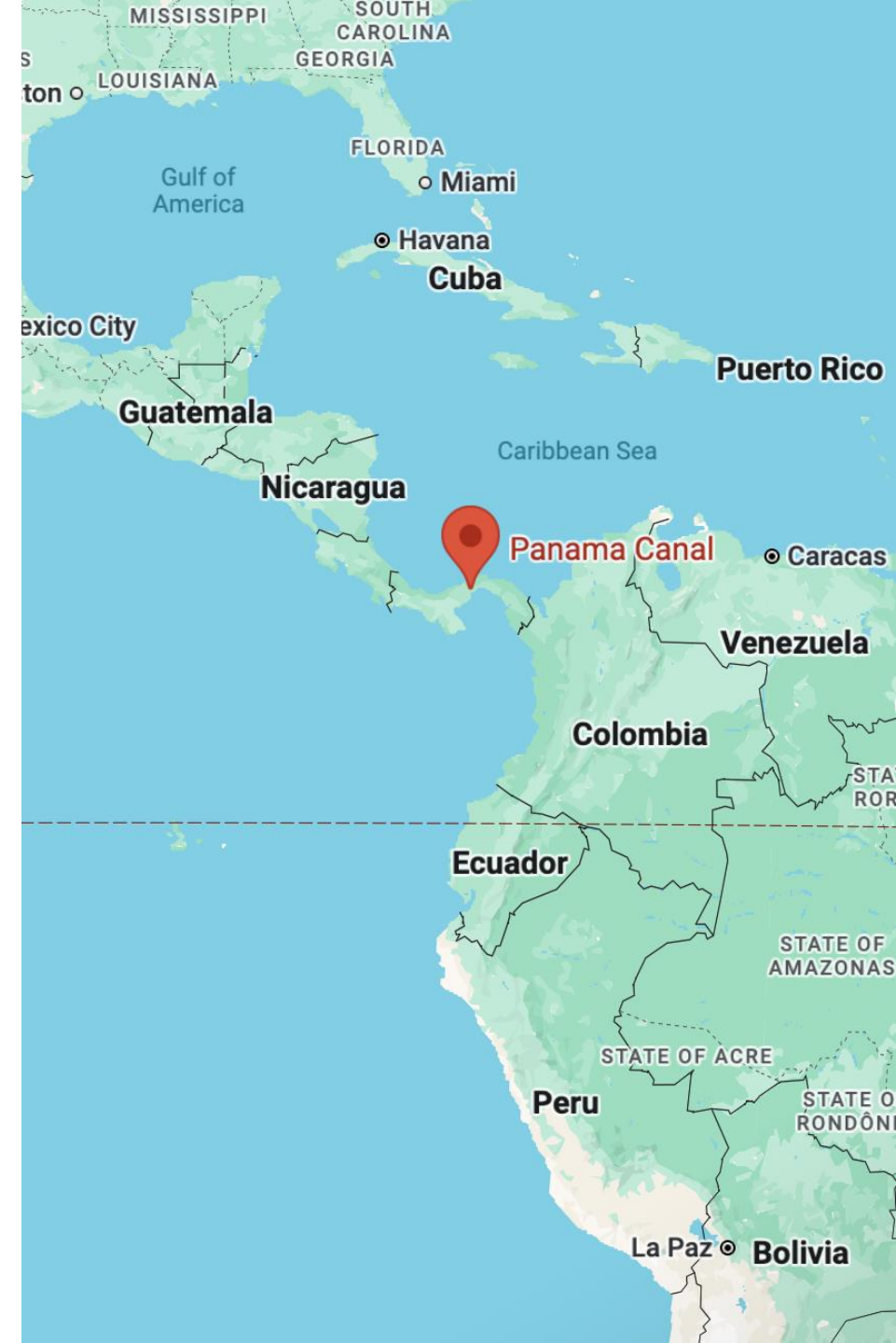
- General Converse

$$\text{"bits to encode } Y" \geq H(Y) \geq I(X; Y) \geq I(X; \hat{X}) \geq R(D)$$

If the "bottleneck" (in bits) is less than $R(D)$, it would be impossible to reconstruct X within distortion D

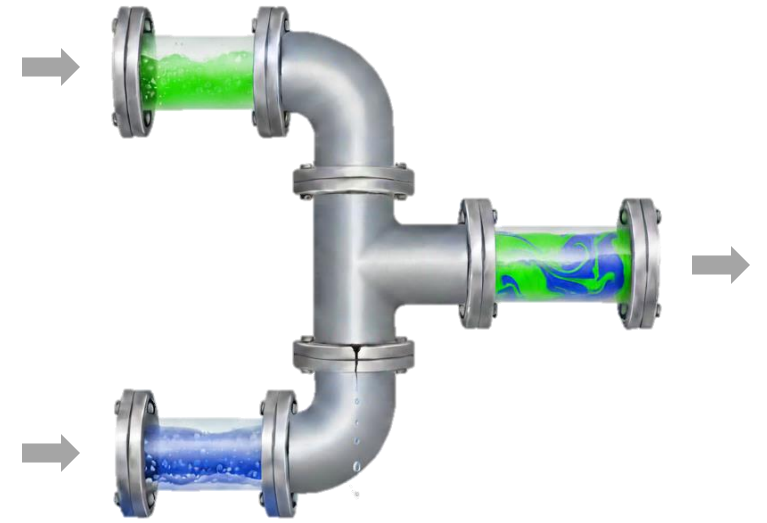


It is the bottleneck that limits the maximum information going through a network.



Takeaways (informal)

- **Information** => “water”
- **Functions** => “pipes”
- “water” never increases in “pipes”
 - if multiple origins: “water” from each origin never increases
- “water” can leak out (which cannot recover)
- narrowest “pipe” (bottleneck) determines the maximum capacity of “water flow”



Modeling Information with Neural Networks

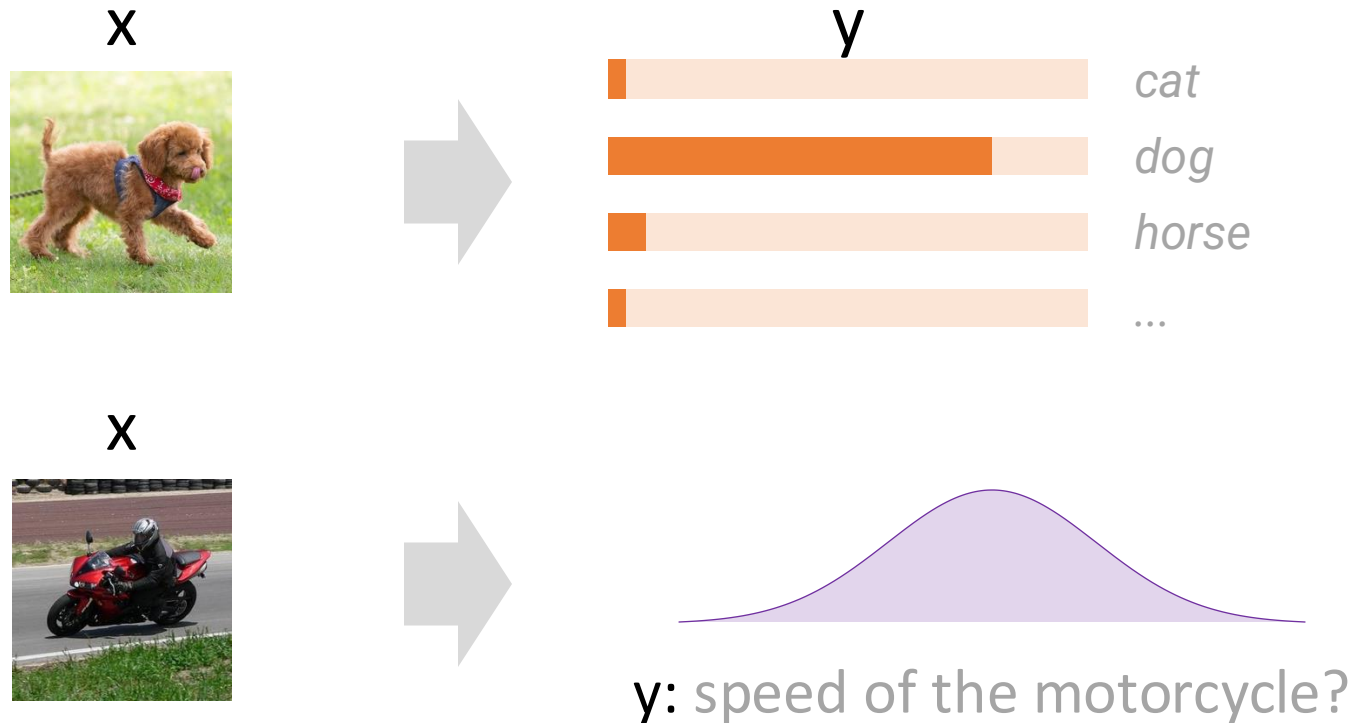
- The foundation of modeling information is the assumption of probability.
- But where does probability come from?
- tl; dr: uncertainty.

Where Does Probability Come From?

- X : an observation (e.g., images)
- Y : an attribute (e.g., class labels)
- Model $P(X, Y)$ as a joint distribution
 - Many plausible observations and attributes.
- Model $P(Y | X)$ as a conditional distribution
 - Given an observation, the attribute is uncertain
- Model $P(X | Y)$ as a conditional distribution
 - Given an attribute, the observation is uncertain

- Model $P(Y | X)$ as a conditional probability
 - Given an observation, the attribute is uncertain

discriminative
models

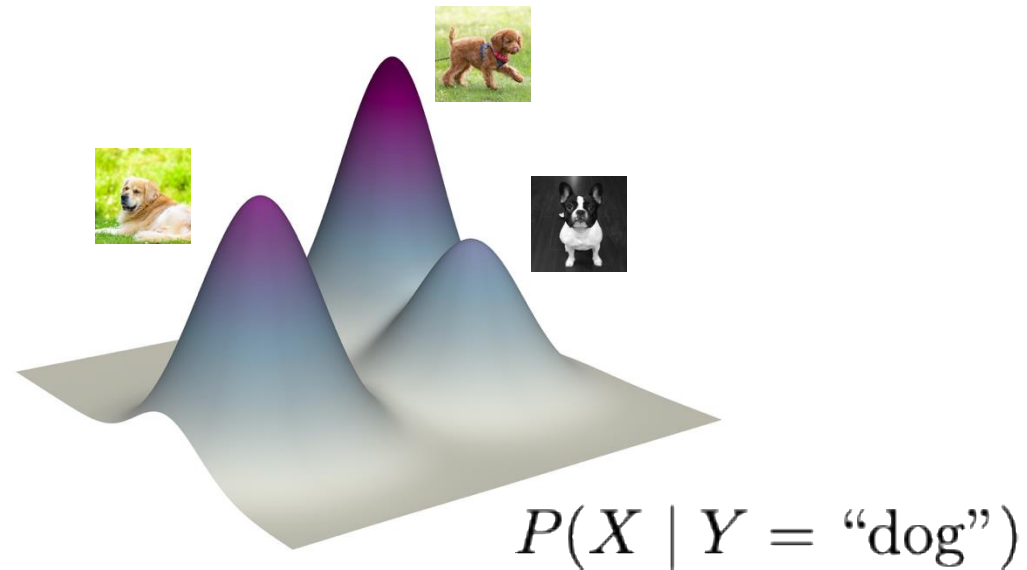


Uncertainty can arise from: noisy labels, imperfect models, or intrinsic ambiguity in the problem itself.

generative
models

- Model $P(X | Y)$ as a conditional probability
 - Given an attribute, the observation is uncertain

$Y = \text{"dog"}$



- The “observation vs. attribute” distinction is **artificial**
- In general, X and Y can be **any variables**



Practical takeaway:
Think about **sources of uncertainty**

[Recap] What are “good” representations?

We may want representations that:

- retain information about the input
- discard information irrelevant to the tasks
- compress information effectively under limited capacity

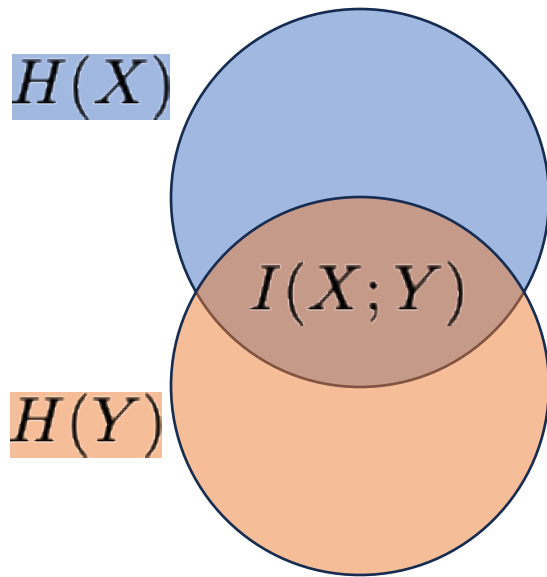
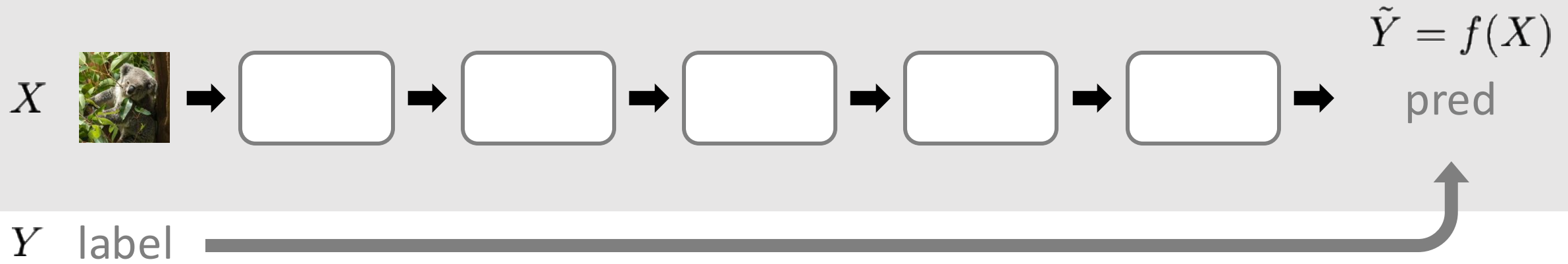
An Information-theoretic Perspective on ...

- Supervised Learning
- Self-supervised Learning: Autoencoding
- Self-supervised Learning: Contrastive Learning

An Information-theoretic Perspective on ...

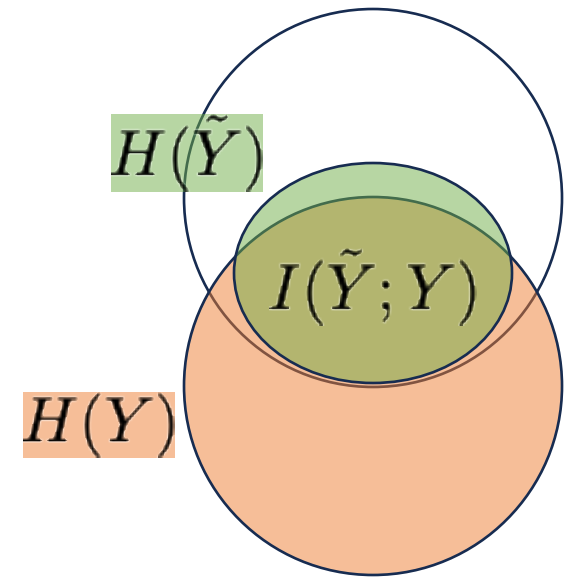
- Supervised Learning
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Supervised Learning

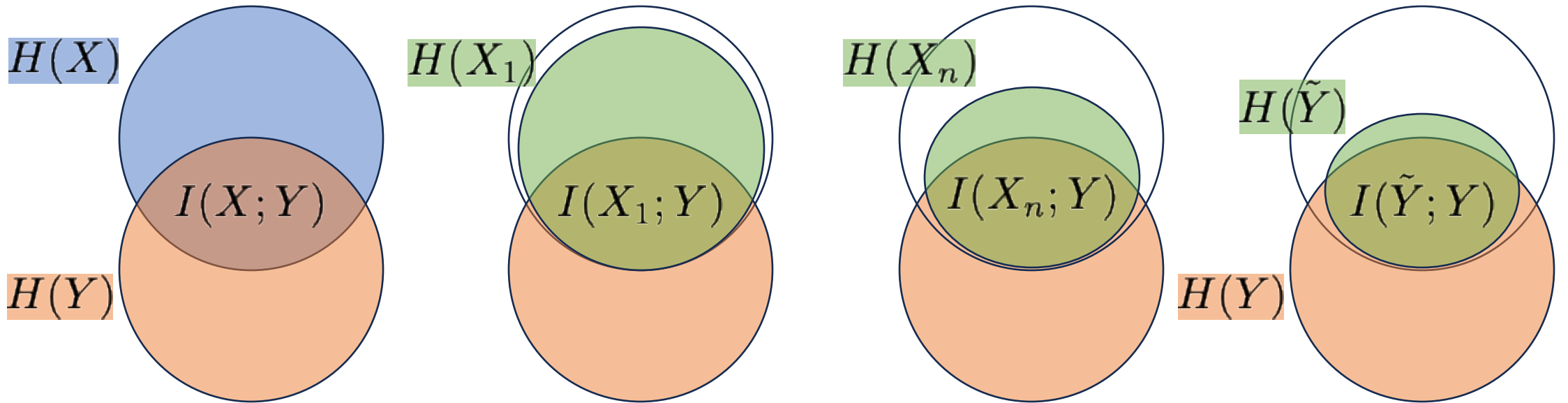
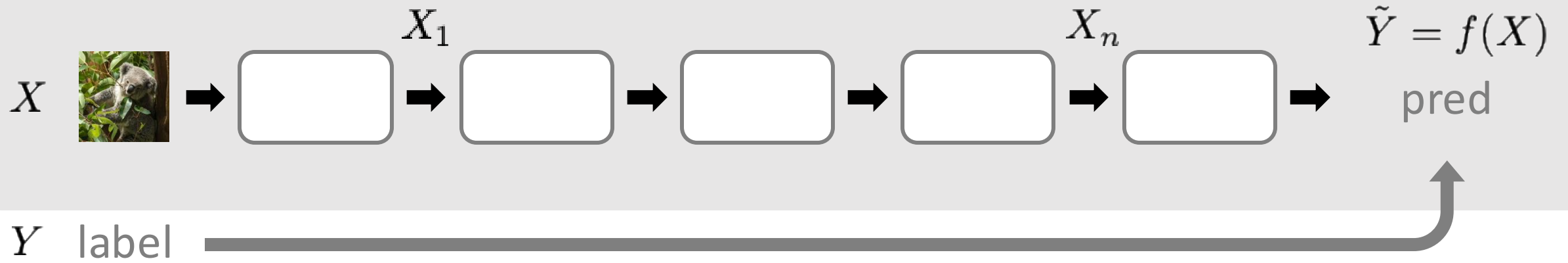


$$I(X; Y) \geq I(\tilde{Y}; Y)$$

(Data Processing Inequality)

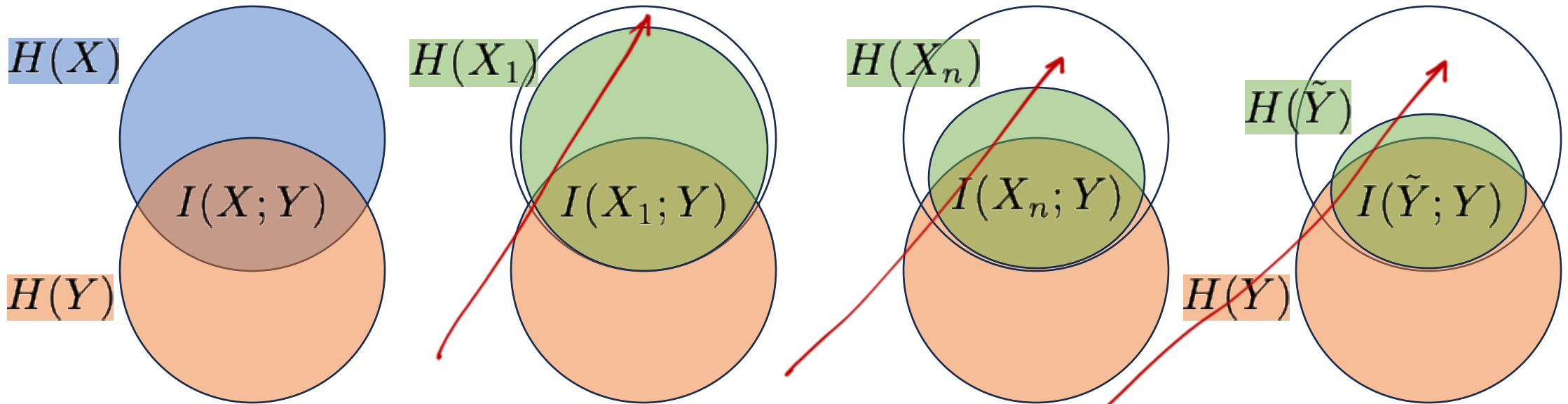
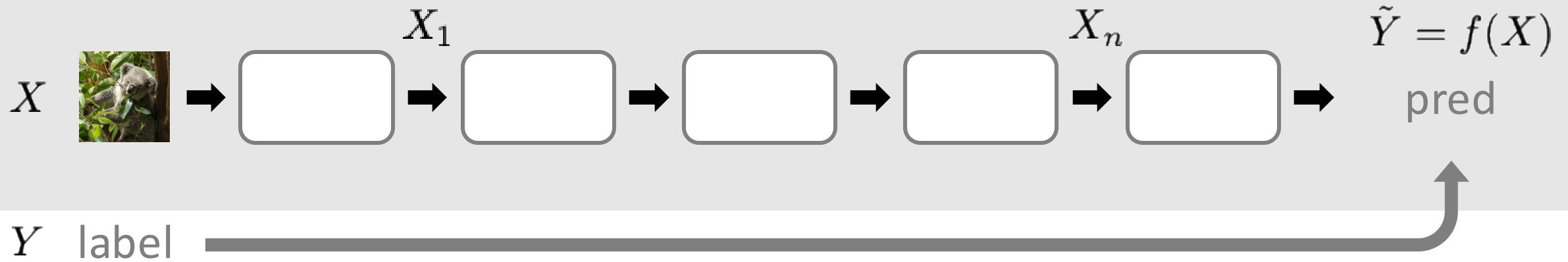


Supervised Learning



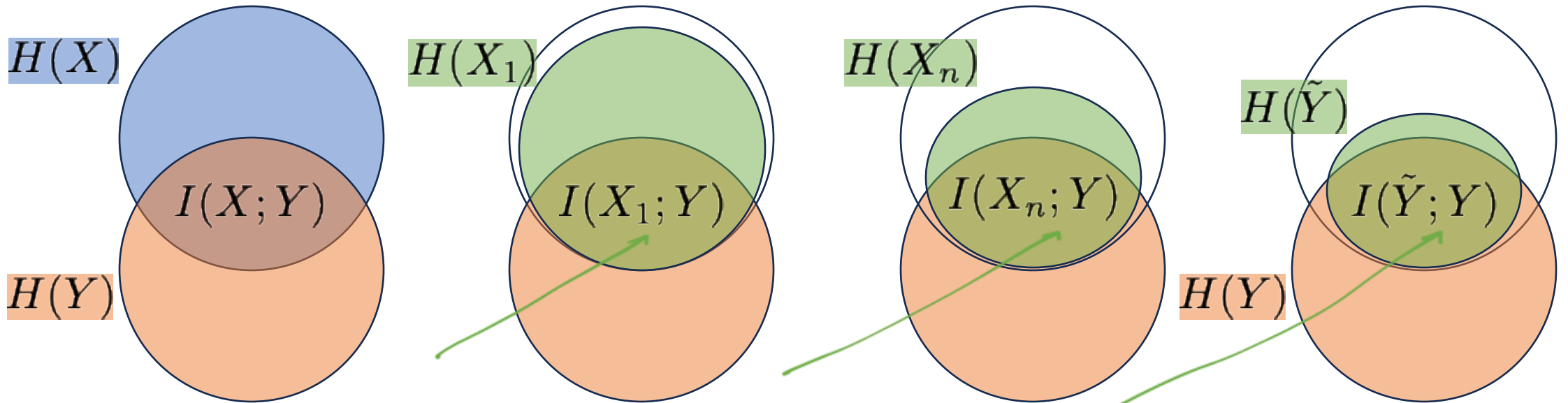
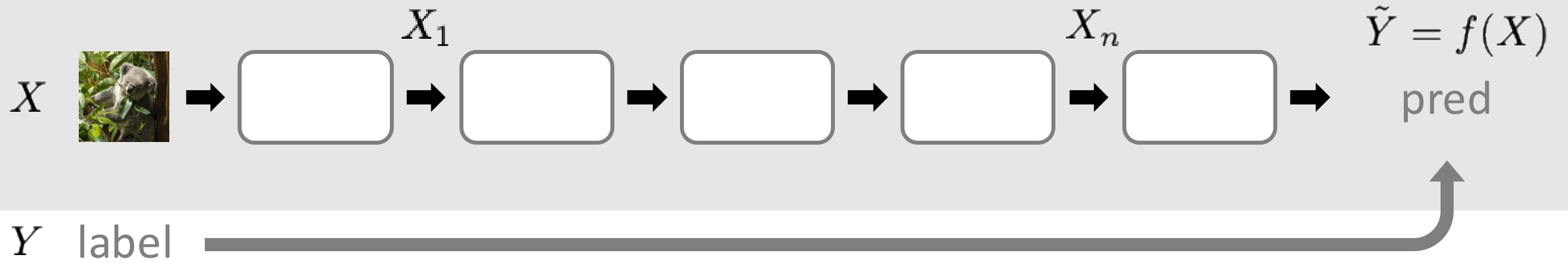
$$I(X; Y) \geq I(X_1; Y) \geq \dots \geq I(X_n; Y) \geq I(\tilde{Y}; Y)$$

Supervised Learning



- **Information discarded:** ideally task-irrelevant

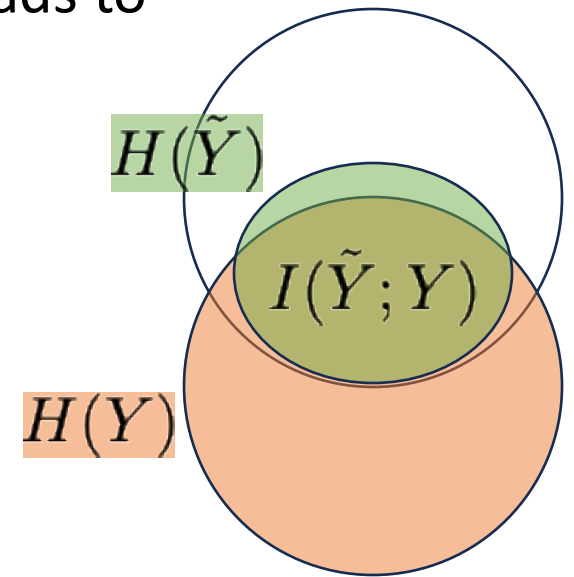
Supervised Learning



- **Information retained:** ideally only about the label Y

In Supervised Learning, ...

- $I(X; Y)$: the target information to retain
- $H(X | Y)$: ideally, the information to discard
- Cross-entropy loss encourages retaining $I(X; Y)$
 - if optimum is achievable, minimal cross-entropy loss leads to maximal mutual information $I(\tilde{Y}; Y) = I(X; Y)$



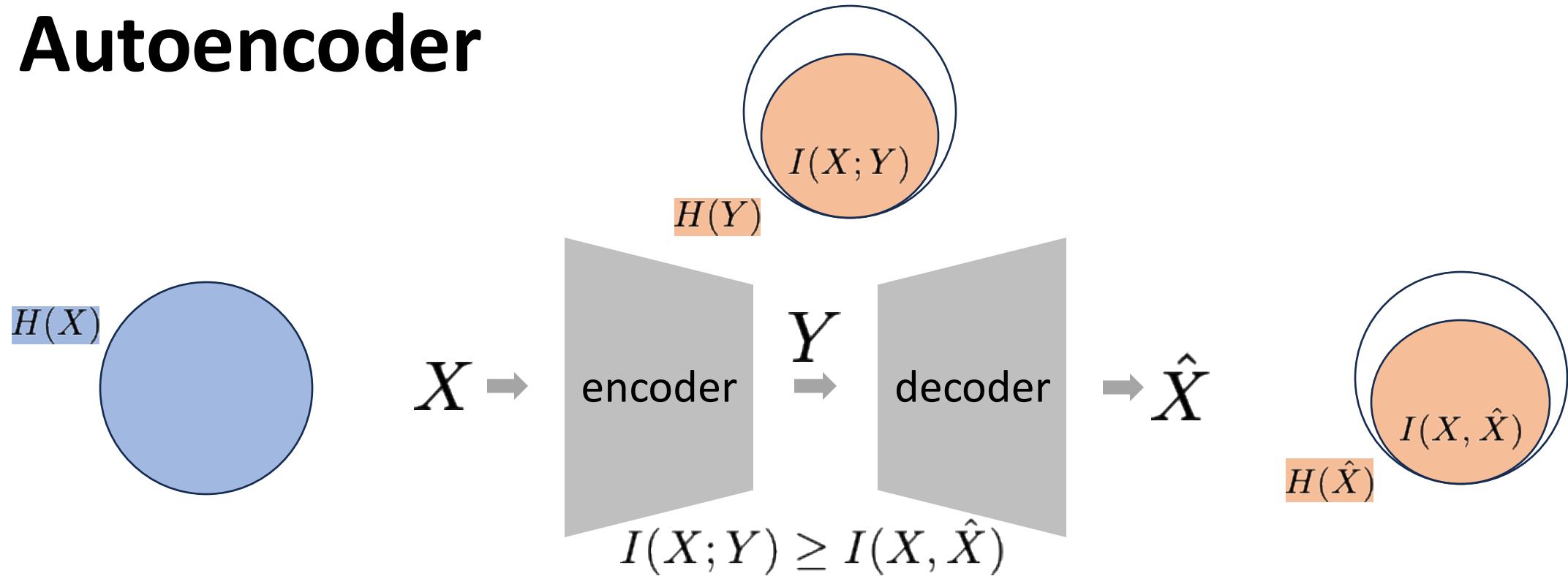
What Makes Modeling Information Difficult?

- Analytical formulations
 - Only for limited families (Gaussian, uniform, categorical, ...)
 - low-dimensional
 - In supervised classification: categorical
- Monte Carlo estimation
 - Distributions exist but unknown
 - We can only sample from the unknown distributions
 - In supervised classification: $x, y \sim p_{\text{data}}(x, y)$
- Information measures
 - often not directly computable (a surrogate is used)
 - In supervised classification: cross-entropy loss as a surrogate of $I(X; Y)$

An Information-theoretic Perspective on ...

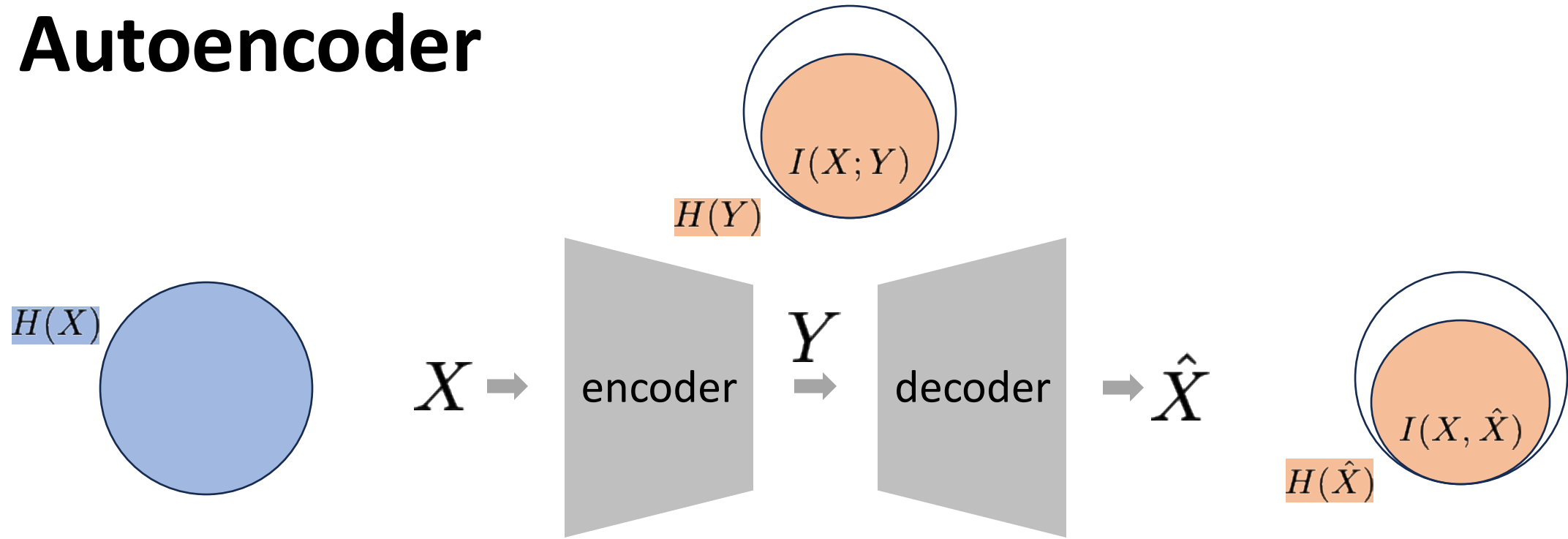
- Supervised Learning
- Self-supervised Learning: Autoencoding
- Self-supervised Learning: Contrastive Learning

Autoencoder



- The bottleneck limits the maximal information in the output
- Even a perfect decoder cannot recover more information (than what is passed through the bottleneck)

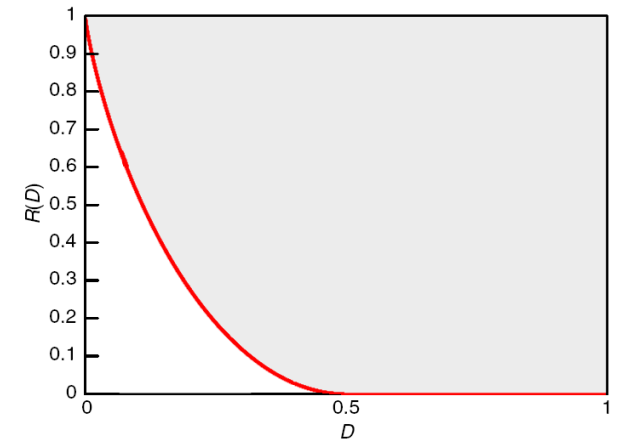
Autoencoder



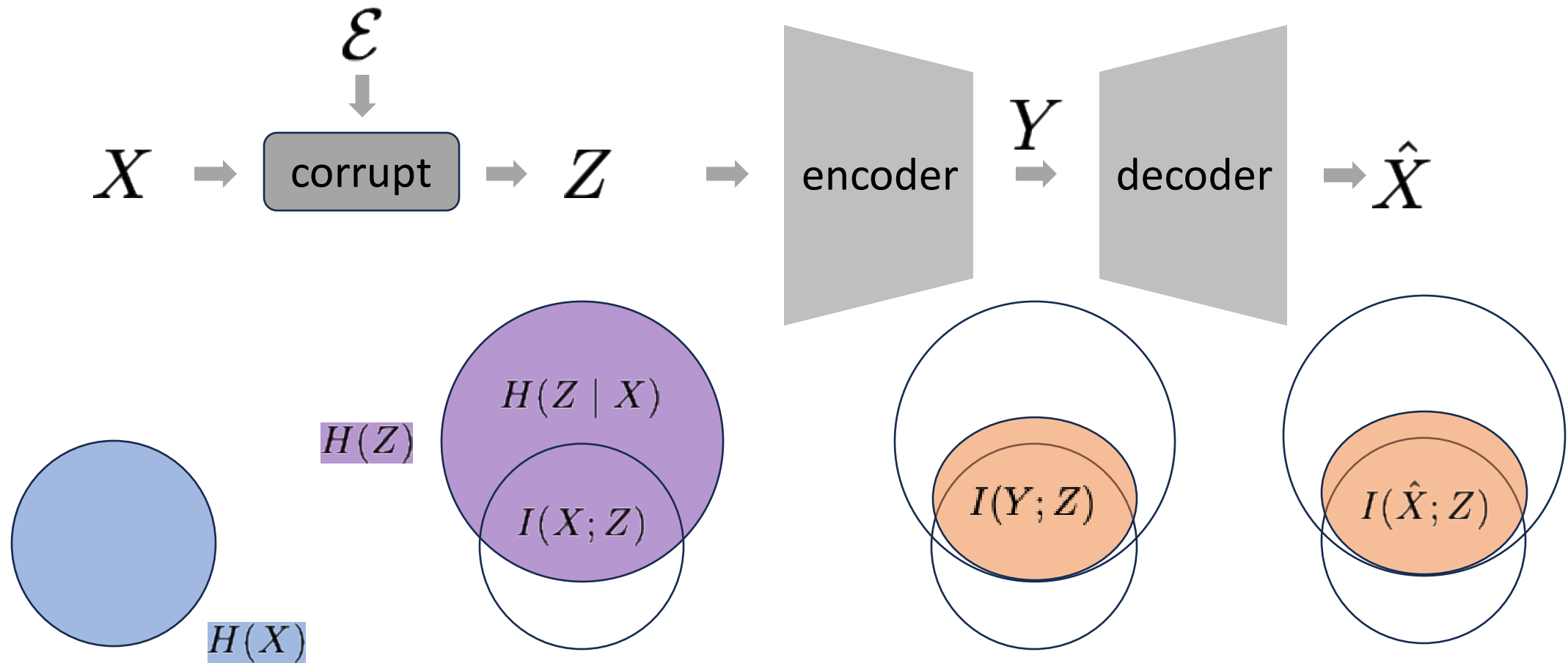
$$\text{"bits to encode } Y" \geq H(Y) \geq I(X; Y) \geq I(X; \hat{X}) \geq R(D)$$

As per Rate-distortion theory:

- Smaller distortion $D \Rightarrow$ larger rate $R(D)$
- Minimizing distortion encourages more information retained
- Reconstruction error acts as a surrogate



Denoising / Masked Autoencoder



- Corruption discards some information
- Encoder can at best retain the remaining mutual information $I(X; Z)$

An Information-theoretic Perspective on ...

- Supervised Learning
- Self-supervised Learning: Autoencoding
- Self-supervised Learning: Contrastive Learning

Mutual Information Neural Estimation (MINE)

Problem setting:

- We have joint distribution $P(X, Y)$
 - unknown, but we can sample from it
- Goal: estimate mutual information $I(X; Y)$
- Approach: learn a neural net $T_\theta(x, y) : X \times Y \rightarrow \mathbb{R}$

Mutual Information:

$$I(X; Y) \triangleq D_{\text{KL}}(\underbrace{p(x, y)}_{\text{joint}} \parallel \underbrace{p(x)p(y)}_{\text{marginals}})$$

Mutual Information Neural Estimation (MINE)

- Donsker-Varadhan's dual representation of KL divergence:

$$D_{\text{KL}}(p \parallel q) = \sup_{T: \mathcal{X} \rightarrow \mathbb{R}} \mathbb{E}_p[T] - \log \mathbb{E}_q[e^T]$$

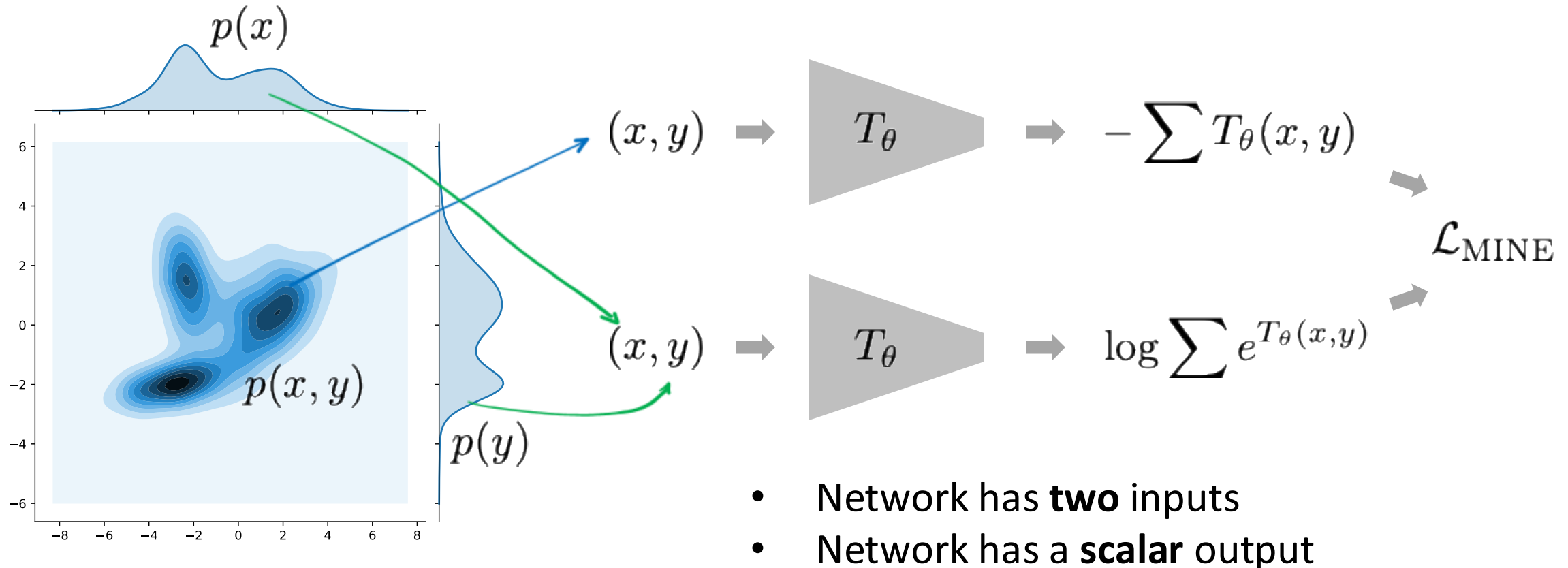
- Mutual Information Neural Estimation (**MINE**):

$$I(X; Y) = D_{\text{KL}}(\underbrace{p(x, y)}_{\text{joint}} \parallel \underbrace{p(x)p(y)}_{\text{marginals}}) \geq \sup_{T_\theta} \underbrace{\mathbb{E}_{p(x, y)}[T_\theta]}_{\text{joint}} - \log \underbrace{\mathbb{E}_{p(x)p(y)}[e^{T_\theta}]}_{\text{marginals}}$$

minimize loss: $\mathcal{L}_{\text{MINE}}(\theta) = -\mathbb{E}_{p(x, y)}[T_\theta] + \log \mathbb{E}_{p(x)p(y)}[e^{T_\theta}]$

Mutual Information Neural Estimation (MINE)

$$\mathcal{L}_{\text{MINE}}(\theta) = -\underbrace{\mathbb{E}_{p(x,y)}[T_\theta]}_{\text{joint}} + \log \underbrace{\mathbb{E}_{p(x)p(y)}[e^{T_\theta}]}_{\text{marginals}}$$



InfoNCE (Noisy Contrastive Estimation)

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{\exp(f(x) \cdot g(y^+) / \tau)}{\sum_{j=0}^N \exp(f(x) \cdot g(y_j) / \tau)} \right]$$

- x : **query** (anchor)
 - y^+ : a “**positive**” sample
 - y_j : candidates (1 “positive”, N “**negatives**”)
 - f, g : networks with **vector** output
 - τ : temperature
- } What are positives and negatives?

InfoNCE (Noisy Contrastive Estimation)

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Relation to MINE:

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[f(x) \cdot g(y^+) / \tau \right] + \mathbb{E} \left[\log \sum_j \exp(f(x) \cdot g(y_j) / \tau) \right]$$

$$\mathcal{L}_{\text{MINE}}(\theta) = -\mathbb{E}_{p(x,y)} [T_\theta] + \log \mathbb{E}_{p(x)p(y)} [e^{T_\theta}]$$

InfoNCE (Noisy Contrastive Estimation)

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Relation to MINE:

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$$\mathcal{L}_{\text{MINE}}(\theta) = -\mathbb{E}_{p(x,y)} [T_\theta] + \log \mathbb{E}_{p(x)p(y)} [e^{T_\theta}]$$

- f, g : accept a single input variable
- $T = f \cdot g / \tau$: restricted family of functions

InfoNCE (Noisy Contrastive Estimation)

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{\exp(f(x) \cdot g(y^+) / \tau)}{\sum_{j=0}^N \exp(f(x) \cdot g(y_j) / \tau)} \right]$$

Relation to MINE:

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[f(x) \cdot g(y^+) / \tau \right] + \mathbb{E} \left[\log \sum_j \exp(f(x) \cdot g(y_j) / \tau) \right]$$

$$\mathcal{L}_{\text{MINE}}(\theta) = -\mathbb{E}_{p(x,y)} [T_\theta] + \log \mathbb{E}_{p(x)p(y)} [e^{T_\theta}]$$

- **InfoNCE:** expectation over finite-sample log-sum-exp
- **MINE:** log of expectation (but approx. by log-sum-exp in practice)

InfoNCE (Noisy Contrastive Estimation)

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{\exp(f(x) \cdot g(y^+) / \tau)}{\sum_{j=0}^N \exp(f(x) \cdot g(y_j) / \tau)} \right]$$

Relation to MINE:

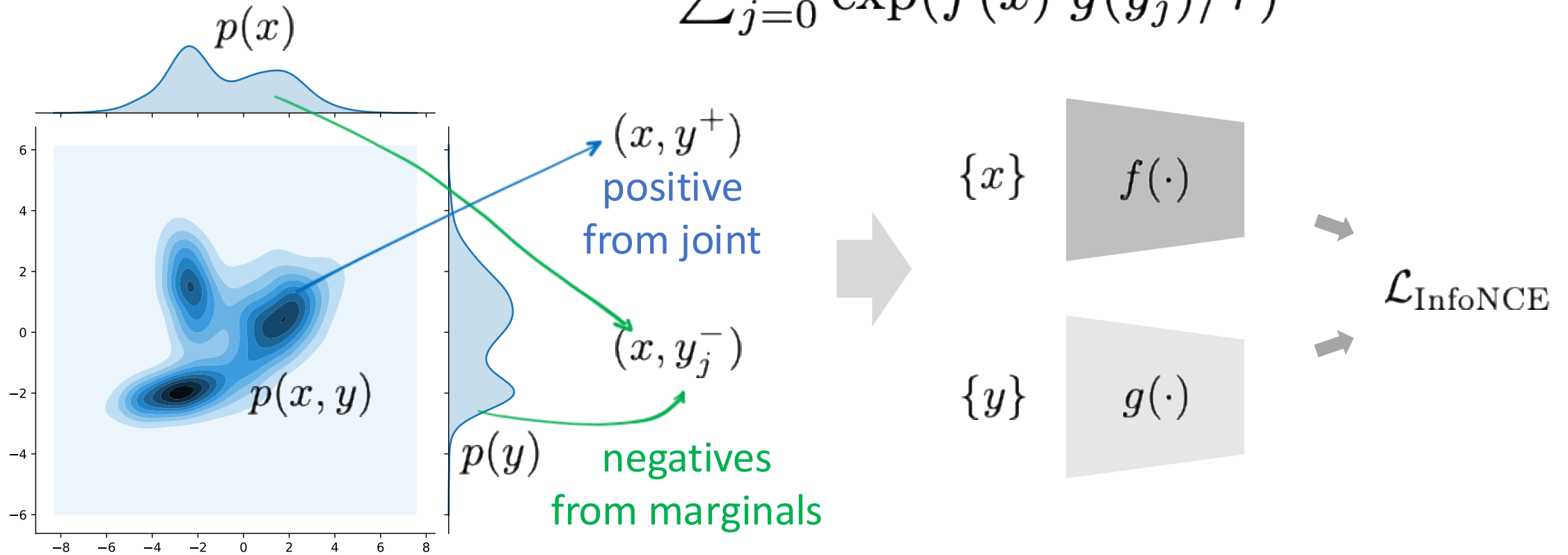
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$$\mathcal{L}_{\text{MINE}}(\theta) = -\mathbb{E}_{p(x,y)} [T_\theta] + \log \mathbb{E}_{p(x)p(y)} [e^{T_\theta}]$$

- “**positive**” pair: from **joint** distribution
- “**negative**” pairs: from product of **marginals**

InfoNCE (Noisy Contrastive Estimation)

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{\exp(f(x) \cdot g(y^+) / \tau)}{\sum_{j=0}^N \exp(f(x) \cdot g(y_j) / \tau)} \right]$$



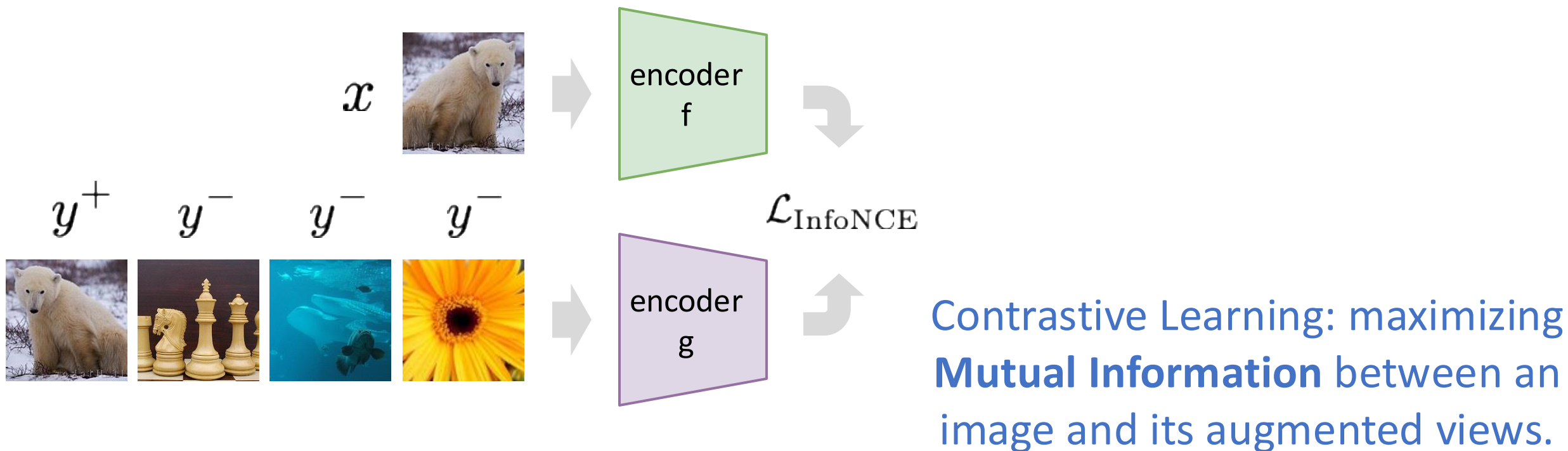
InfoNCE (Noisy Contrastive Estimation)

Estimating Mutual Information via InfoNCE:

$$I(X; Y) \geq \log(N) - \mathcal{L}_{\text{InfoNCE}}$$

- $\mathcal{L}_{\text{InfoNCE}}$ provides a lower bound on Mutual Information (up to a known constant $\log(N)$)
- Minimizing $\mathcal{L}_{\text{InfoNCE}}$ is to maximize the lower bound
- Bound tightens as N increases

Case Study: Contrastive Learning



- $x \sim p_{\text{im}}$, $y \sim p_{\text{im}}$: image distribution
- joint: $p(x, y^+) = p(y^+ | x)p(x)$ where $p(y^+ | x)$ is augmentation
- margins: $p(x)p(y)$, two random images

[Re-Recap] What are “good” representations?

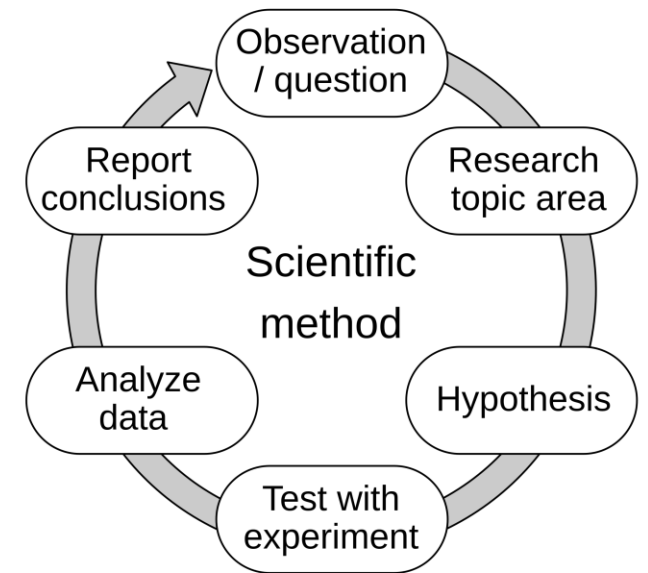
We may want to answer:

- retain what information about the input?
- discard what information irrelevant to the tasks?
- compress what information under limited capacity?

still open questions ...

A note before we go ...

- Deep learning research is largely **empirical**
- Problem solving by **trial-and-error**
 - “human gradient descent”
 - with “reward” from experiments
- Theories provide good **motivations**
 - but evidence is needed



References

- Lecture Notes, 6.7480, 6.441, MIT. Y Polyanskiy
- *“Information Theory and Network Coding”*, R Yeung. Springer, 2008
- *“The information bottleneck method”*, Tishby et al., 2000
- *“Extracting and Composing Robust Features with Denoising Autoencoders”*, Vincent et al., ICML, 2008
- *“Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion”*, Vincent et al., JMLR, 2010
- *“Mutual Information Neural Estimation”*, Belghazi et al., ICML 2018
- *“Representation Learning with Contrastive Predictive Coding”*, van den Oord et al., arXiv 2018
- *“Learning deep representations by mutual information estimation and maximization”*, R Hjelm et al., ICLR 2019

Overview: This Lecture

- Basics of Information Theory
- Modeling Information with Neural Networks